Convolution as a local operation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make every output traceable to a patch and contract.

Show all products and complete maps;separate shared parameter slots,operation counts,boundary assumptions and activation.

Keep one habit: Assess independent products, shape, bias and ReLU with checked keys.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Compute a local weighted sum at each valid image position

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

The same dimensions can describe different coordinate meanings. A correct computation must know which axis is row, column or channel.

Opening retrieval

Without opening the old lesson, explain one key idea from Image tensors and channels. Give an example and a mistake that the idea helps you avoid.

Look for: Read a pixel using declared row column and channel axes

Reference example, not the only acceptable wording: Teacher explanation: The RGB pixel is [0, 0, 255]; its blue value is 255. Shape is 2 by 2 by 3, with four pixels and twelve channel values.

Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

Look for: Check dimension contracts before computing a layer

Reference example, not the only acceptable wording: Teacher explanation: No. Each weight row must have the same length as x.

Confirm the intended feature schema, then supply a valid two-feature **input** or deliberately redesign and retrain the layer; do not silently discard a value.

Without opening the old lesson, explain one key idea from **Training** and validation curves. Give an example and a mistake that the idea helps you avoid.

Look for: Compute comparable **training** and validation losses at saved checkpoints

Reference example, not the only acceptable wording: Teacher explanation: **Prediction**s [3, 4], residuals [-1.5, -2], squares [2.25, 4], MSE 3.125.

Pictured end goal

By the end, I can explain convolution as a local operation using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Image tensors and channels

Debugging shapes and gradients

Training and validation curves

For the compact alternative schedule, use two 45-minute sessions. Prepare a 2 by 2 paper window and a labelled 3 by 3 grid. Keep **kernel** orientation fixed and distinguish raw sums, bias and ReLU.

Explain that neural-network convolution commonly uses cross-correlation; the exercise explicitly declares this convention. Technical reference: https://docs.pytorch.org/docs/stable/generated/torch.nn.modules.conv.Conv2d.html .

No library installation is needed.

Outcomes

Compute a local weighted sum at each valid image position

Explain shared weights and predict **output** dimensions

Check how **stride** boundaries and activation affect the result

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach exact patch arithmetic under a named boundary convention

> Keep main,pictured and independent arrays explicitly separate.

Main contract, complete counts and guided key

Use a fictional single-channel image with rows [1, 2, 0], [0, 1, 3] and [2, 0, 1]. Coordinates are zero-based row then column. Our **kernel** has rows [1, -1] and [0, 1].

At each position, multiply matching entries of a 2 by 2 window and the **kernel**, then add the four products. Use bias 0, **stride** 1 and no **padding**.

At the top-left window, the sum is 1 times 1 plus 2 times -1 plus 0 times 0 plus 1 times 1 = 0. Slide the same **kernel** right, then down. Only use windows that fit entirely inside the image; this is the valid boundary rule.

The **feature map** has rows [0, 5] and [-1, -1]. This neural-network convention keeps the **kernel** orientation fixed; mathematically it is cross-correlation. Mathematical convolution reverses the **kernel** along both spatial axes.

We use the fixed-orientation convention throughout, so every hand calculation and program agrees.

The following function assumes nonempty rectangular numeric arrays and a positive integer **stride**. It rejects a **kernel** larger than the image.

It computes one **input** channel, one **output** map, zero bias and no **padding**; no activation is applied inside the function.

The **kernel** weights are reused at all four positions. There are four **kernel** parameters, not sixteen separate weights. If we add one shared bias for this **output** map, there are five parameters.

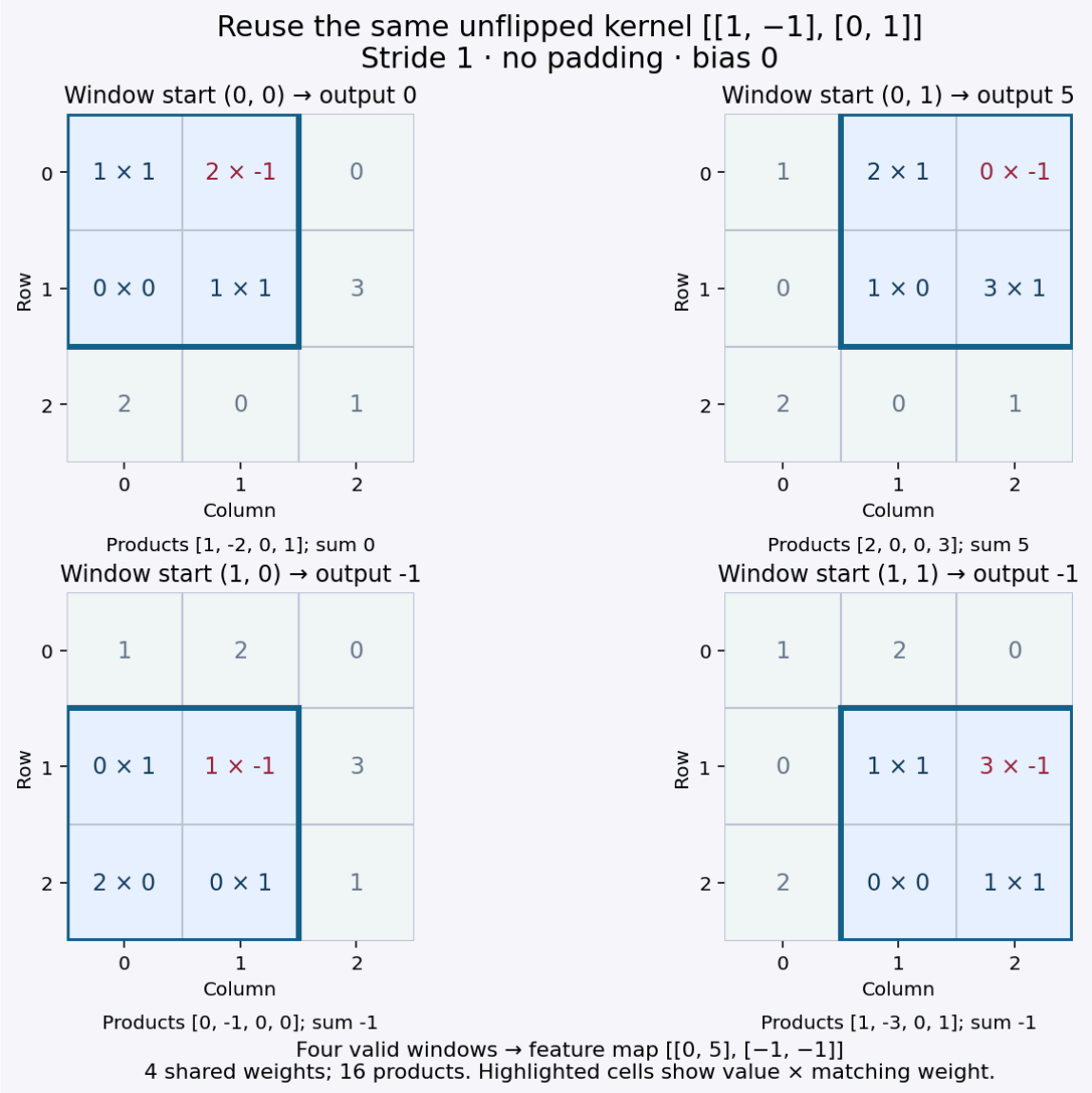
The number of multiply operations grows with the number of windows even though the parameter count stays fixed.

Main local operation	Complete count / meaning
Input / kernel / output	3×3 image, 2×2 kernel, 2×2 feature map

Main local operation	Complete count / meaning
Allowed starts with stride 1	Rows0/1 and columns 0/1:four windows
Declared shared kernel values	Four weights reused at all four starts
Products in this direct algorithm	Four per window×four windows=16, including products with weight 0
Add one shared bias	Five parameter slots; add that bias once per output
Activation	No activation inside local_filter; apply a declared ReLU afterward if needed
Status	Hand-chosen arithmetic kernel, not a fitted recognition model or measured hardware-operation benchmark

Guided: only the window starting [0, 0] includes **input**[0][0]. Its weight is 1, so adding 3 changes that **output** from 0 to 3. The map becomes [[3, 5], [-1, -1]]. Exit: the declared unpadded **stride** 1 setup gives 2 by 2, not 3 by 3. A different **padding** or **kernel** setup would need documentation. Values are local weighted sums, not probabilities.

All four weighted windows and one-factor variants



One-factor main-image variant	Complete output / boundary meaning
Original stride 1, no padding, bias 0	[[0, 5], [−1, −1]], shape 2×2
Stride2, same image/kernel	Only start(0, 0); map[[0]], shape 1×1
One zero border per side, stride 1	Padded5×5 input→4×4 map; top-left window[[0, 0], [0, 1]]gives 1
ReLU after original raw map	[[0, 5], [0, 0]]; raw negative sums remain valid debugging evidence

One-factor main-image variant	Complete output / boundary meaning
Reverse both axes of kernel	[[1, 0], [−1, 1]]is a changed convention; top-left result 2 rather than 0
Change only input[0][0]1→4	[[3, 5], [−1, −1]]; only first valid window contains that corner

Complete separate pictured arrays and scope

Draw the exact patch-by-patch arithmetic of a local image operation.

Image rows [1, 2, 3],[4, 5, 6]; **kernel** rows [1, 0],[0,-1]. Use unflipped **kernel**, no **padding**, **stride** 1, bias 0.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses a different 2×3 image[[1, 2, 3], [4, 5, 6]]and unflipped kernel[[1, 0], [0, −1]], stride 1, no padding, bias 0. Its two windows give 1−5=−4 and2−6=−4, map[[-4, −4]]. Adding only shared bias 2 gives[[-2, −2]]without changing products, weight sharing or1×2 shape. These negative raw responses are allowed. The main 3×3 image instead gives[[0, 5], [−1, −1]].

Keep the image and kernel beside each calculation.

Trace unchanged products under a shared bias addition

AI ACADEMY / AI-5/21 · 02 · TRACE THE CASE

Follow the deciding steps

01 First patch

1×1+2×0+4×0+5×(-1)=-4.

02 Second patch

2×1+3×0+5×0+6×(-1)=-4.

03 Feature map

Output shape 1×2 and map [[-4,-4]]; the same kernel is reused at each position.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. First patch	1×1+2×0+4×0+5×(-1)=-4.
2. Second patch	2×1+3×0+5×0+6×(-1)=-4.
3. Feature map	Output shape 1×2 and map [[-4,-4]]; the same kernel is reused at each position.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A local **kernel** combines the values in one small image window with a fixed set of weights. Slide that same **kernel** to the next valid window and repeat the products.

Sharing weights lets the operation look for the same local pattern at different positions. This example declares an unflipped **kernel**; adding bias changes each window result without changing which pixels were used.

A local filter detects a numerical pattern; one **feature map** does not establish recognition or learned visual understanding.

Change: Add shared bias 2 after each patch sum.

Prediction: **Feature map** becomes [[-2,-2]].

Reason: Bias adds 2 to each **output**; it does not change patch products or **output** shape.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Kernel	A small array of weights applied to a local input neighbourhood
Feature map	The spatial array of outputs produced by applying a filter
Stride	The distance between successive window starting positions
Padding	Values added around an input boundary under a declared rule

One shared kernel produces four local sums

WINDOW START	KERNEL ROWS	OUTPUT VALUE
Row 0 column 0	[1 -1] and [0 1]	0
Row 0 column 1	[1 -1] and [0 1]	5
Row 1 column 0	[1 -1] and [0 1]	-1
Row 1 column 1	[1 -1] and [0 1]	-1

Read windows in row order. Each output summarizes one neighbourhood. With stride 1, adjacent windows overlap, but their output values occupy separate cells.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

One shared kernel produces four local sums

WINDOW START	KERNEL ROWS	OUTPUT VALUE
Row 0 column 0	[1 -1] and [0 1]	0
Row 0 column 1	[1 -1] and [0 1]	5
Row 1 column 0	[1 -1] and [0 1]	-1
Row 1 column 1	[1 -1] and [0 1]	-1

Read windows in row order. Each output summarizes one neighbourhood. With stride 1, adjacent windows overlap, but their output values occupy separate cells.

Change only input[0][0] from 1 to 4. Which valid windows include it? Compute the changed feature map and explain why the other three outputs stay the same.

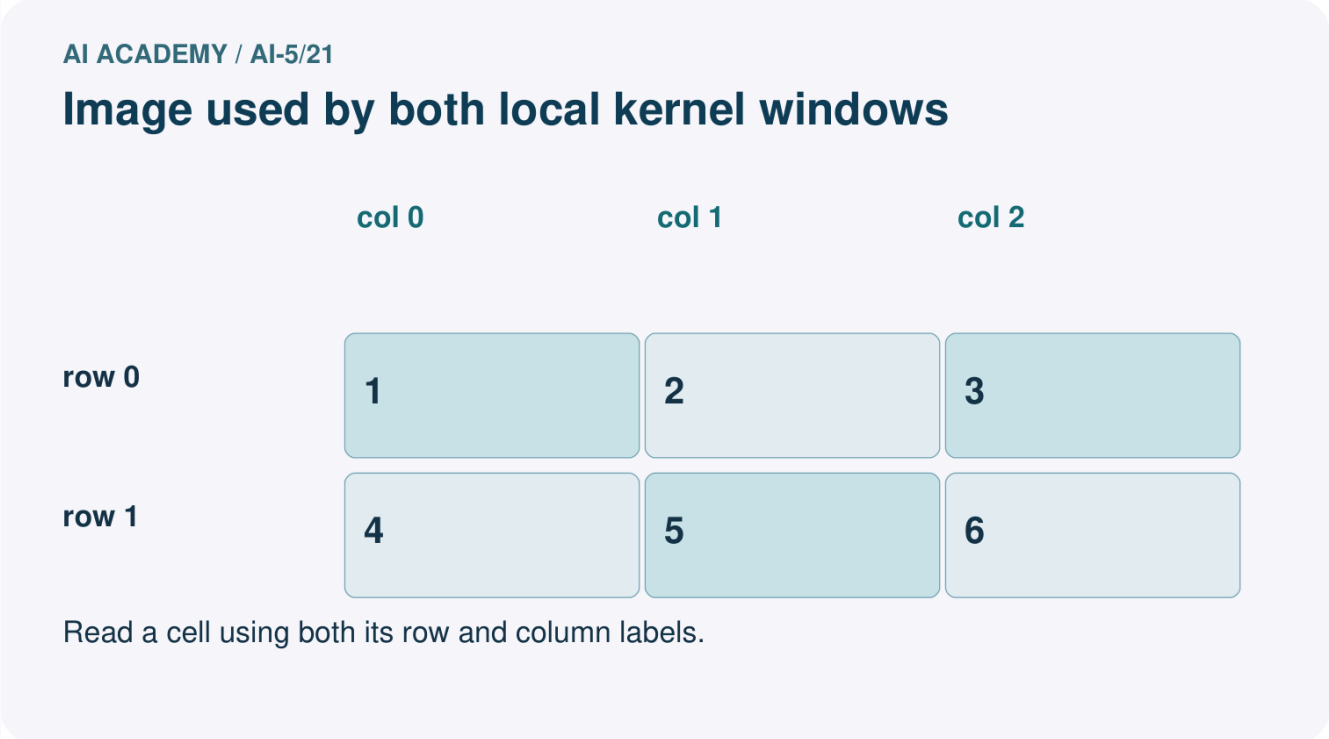
Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move	Teacher prompt
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Convolution as a local operation

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/21

Convolution as a local operation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First patch

1 x 1+2 x 0+4 x 0+5 x (-1)=-4.



2. Second patch

2 x 1+3 x 0+5 x 0+6 x (-1)=-4.



3. Feature map

Output shape 1 x 2 and map [[-4,-4]]; the same kernel is reused at each position.

CHANGE ONE THING

Add shared bias 2 after each patch sum.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: How many shared **kernel** weights does the lesson use? How many products are computed over all four valid windows? What changes if one shared bias is added?

Teacher check: Teacher explanation: There are four shared weights and sixteen products across four windows. One shared bias raises the parameter count to five and adds the same bias once to each **output**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Convolution as a local operation. Proposed explanation: “Each window has its own set of learned weights”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The same **kernel** parameters are reused at every position for this **output** map.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Why is flipping the **kernel** an implementation change here? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

The declared operation uses an unflipped **kernel**, often called cross-correlation. Flipping it changes the arithmetic contract and may change results. Changed-case check: **Feature map** becomes [[-2,-2]].

Bias adds 2 to each **output**; it does not change patch products or **output** shape.

Changed-case reasoning: **Feature map** becomes [[-2,-2]]. Bias adds 2 to each **output**; it does not change patch products or **output** shape.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a four-cell product ledger and move a paper window one position at a time. Use labelled coordinates rather than colour alone.

Extension: compare a numerical implementation with an independent explicit four-term formula across rectangular **input**s, and show how changing **stride** alters the sampled start positions.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve row and column axes, then label the input and fixed-orientation kernel.
7 to 18 minutes	Compute the four products at the top-left position and complete W1 at the next window.
18 to 32 minutes	Trace the remaining windows and run or hand-trace the optional code.
32 to 45 minutes	Use W2 to distinguish shared parameter count from repeated arithmetic work.
Session 2 start to 8 minutes	Complete the guided changed-pixel case and explain its local influence.
8 to 20 minutes	Count stride positions and padded windows, then complete W3.
20 to 35 minutes	Collect W4 independently, requiring separate raw, biased and activated maps.
35 to 45 minutes	Use W5 and the exit response to check conventions and claims, then administer the fresh retry.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main four-window products, full padded map and contract supplied. W2 fourweights/sixteenproducts; sharedbiasaddsoneparameter andoneadditionperoutput. W3 stride2 shape1×1; zero-borderstride 1 shape4×4, top-left 1. W5 flippedkernel top-left 2 not0. Independentimage[[2, 1, 0], [1, 3, 2]], kernel[[0, 1], [–1, 2]], validstride 1 bias0:products 0/1/–1/6 sum6 and0/0/–3/4 sum1, map[[6, 1]], shape 1×2.

Addbias–5→[[1, –4]], thenReLU[[1, 0]]. Negative raw responses allowed; none of these numbers is a probability.

Diagnose the current misconceptions

Each window has its own set of learned weights: The same **kernel** parameters are reused at every position for this **output** map.

A negative feature-map value means the code failed: Signed weighted sums are valid; an activation may later change them.

A local response proves an object was recognised: The authored filter only computes a local sum; object recognition and probability claims require additional modelling and **evidence**.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Use image rows [2, 1, 0], [1, 3, 2] and **kernel** rows [0, 1], [-1, 2]. Use no **padding**, **stride** 1 and bias 0. Give the **output** shape, both four-product sums and the **feature map**.

Then add a shared bias of -5 and apply ReLU.

Teacher explanation: Shape is 1 by 2. First products are 0, 1, -1, 6, summing to 6. Second products are 0, 0, -3, 4, summing to 1. Map is [[6, 1]]. Bias gives [[1, -4]]; ReLU gives [[1, 0]].

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trace a local sum

Use the lesson image and **kernel** at window start [0, 1]. List the window rows, four products and **output** before activation.

Teacher explanation: Window rows are [2, 0] and [1, 3]. Products are 2, 0, 0 and 3, giving **output** 5.

W2 Separate parameters and operations

How many shared **kernel** weights does the lesson use? How many products are computed over all four valid windows? What changes if one shared bias is added?

Teacher explanation: There are four shared weights and sixteen products across four windows. One shared bias raises the parameter count to five and adds the same bias once to each **output**.

W3 Check stride and padding

For the lesson image and **kernel**, give the **output** shape with no **padding** and **stride** 2. Then give the shape with one zero border on each side and **stride** 1, and calculate that padded map's top-left **output**.

Teacher explanation: **Stride** 2 without **padding** gives shape 1 by 1. The padded **input** is 5 by 5, giving **output** 4 by 4. Its top-left window [[0, 0], [0, 1]] gives **output** 1.

W5 Test a convention error

A program reverses both axes of the lesson **kernel** to [[1, 0], [-1, 1]]. Compute the top-left **output**. Does it agree with our declared convention? Explain what the test checks.

Teacher explanation: The top-left **output** becomes 1 + 0 + 0 + 1 = 2 instead of 0. It does not follow the declared fixed-orientation convention. The asymmetric **kernel** and known window expose a **kernel**-orientation mismatch.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: only the window starting [0, 0] includes **input**[0][0]. Its weight is 1, so adding 3 changes that **output** from 0 to 3. The map becomes [[3, 5], [-1, -1]]. Exit: the declared unpadded **stride** 1 setup gives 2 by 2, not 3 by 3. A different **padding** or **kernel** setup would need documentation. Values are local weighted sums, not probabilities.

W1 Trace a local sum

W2 Separate parameters and operations

W3 Check stride and padding

W4 Filter a new image independently

Use image rows [2, 1, 0], [1, 3, 2] and **kernel** rows [0, 1], [-1, 2]. Use no **padding**, **stride** 1 and bias 0. Give the **output** shape, both four-product sums and the **feature map**. Then add a shared bias of -5 and apply ReLU.

Teacher explanation: Shape is 1 by 2. First products are 0, 1, -1, 6, summing to 6. Second products are 0, 0, -3, 4, summing to 1. Map is [[6, 1]]. Bias gives [[1, -4]]; ReLU gives [[1, 0]].

W5 Test a convention error

Independent assessment and fresh retry

W4 requires shape 1 by 2, raw **output**s 6 and 1, biased **output**s 1 and -4 and ReLU **output**s 1 and 0. Fresh retry: image rows [1, 2, 4], [3, 0, 1]; **kernel** rows [1, 0], [-1, 1].

Products at the two starts are [1, 0, -3, 0] and [2, 0, 0, 1], giving [[-2, 3]]. Add shared bias 1 to get [[-1, 4]], then ReLU gives [[0, 4]].

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled convolution as a local operation **evidence** sheet to Image-pattern Network Investigation. Show how you can compute a local weighted sum at each valid image position.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide a four-cell product ledger and move a paper window one position at a time. Use labelled coordinates rather than colour alone.

Extension: compare a numerical implementation with an independent explicit four-term formula across rectangular **input**s, and show how changing **stride** alters the sampled start positions.

Week 22 explores pooling and spatial information, including distinct **input**s that share a pooled **output**.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/22, ai-5/24, ai-5/27. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	

Record	Teacher evidence
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Compute a local weighted sum at each valid image position

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: How many shared **kernel** weights does the lesson use? How many products are computed over all four valid windows? What changes if one shared bias is added?

Reasoned practice answer: Teacher explanation: There are four shared weights and sixteen products across four windows. One shared bias raises the parameter count to five and adds the same bias once to each **output**.

Check every learner before discussion: Use the worked case for Convolution as a local operation. Proposed explanation: “Each window has its own set of learned weights”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The same **kernel** parameters are reused at every position for this **output** map.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Add shared bias 2 after each patch sum. Require a **prediction** and a reason before comparing. Expected result: **Feature map** becomes [[-2,-2]].

Reason: Bias adds 2 to each **output**; it does not change patch products or **output** shape.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

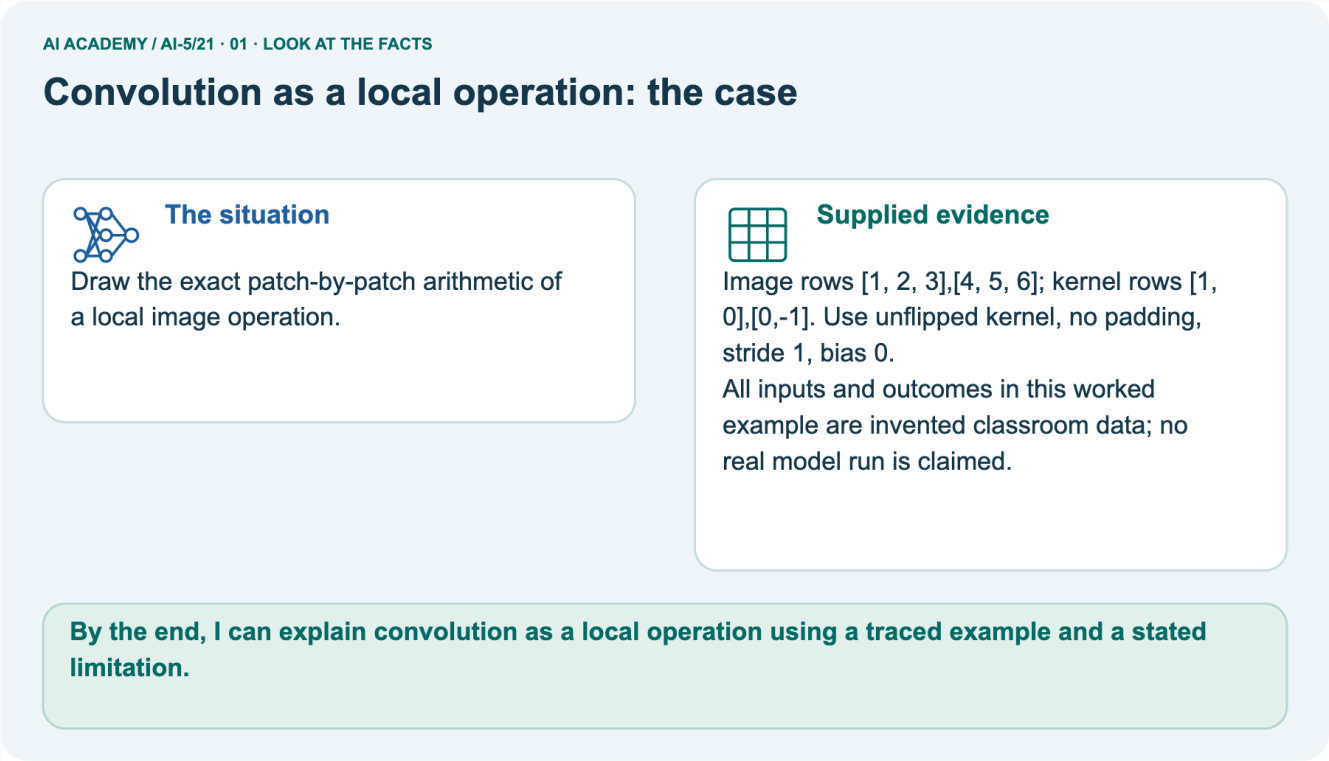
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Convolution as a local operation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

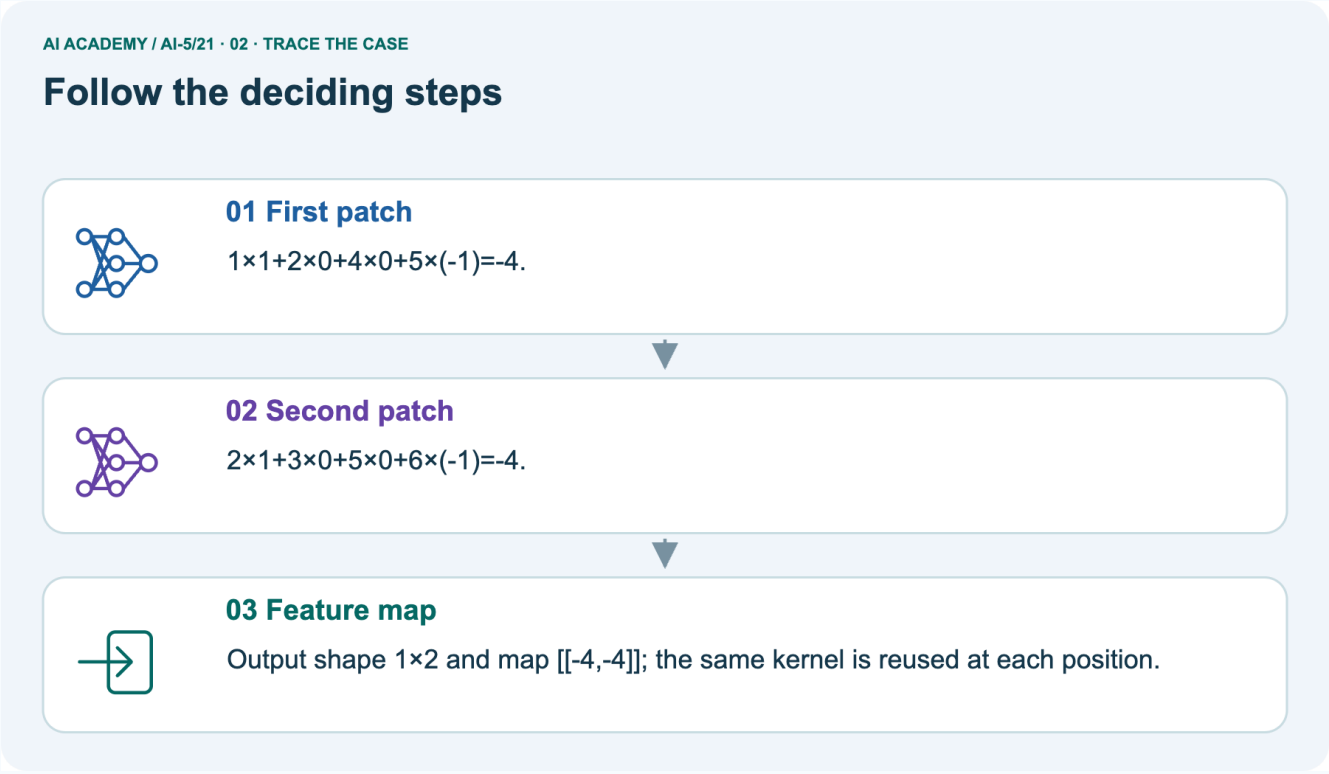
Draw the exact patch-by-patch arithmetic of a local image operation. Supplied facts: Image rows [1, 2, 3],[4, 5, 6]; **kernel** rows [1, 0],[0,-1]. Use unflipped **kernel**, no **padding**, **stride** 1, bias 0.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

First patch: 1×1+2×0+4×0+5×(-1)=-4. → Second patch: 2×1+3×0+5×0+6×(-1)=-4. → **Feature map**: **Output** shape 1×2 and map [[-4,-4]]; the same **kernel** is reused at each position.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Image used by both local kernel windows



Follow the labelled picture with the matching explanation. Open the vector diagram

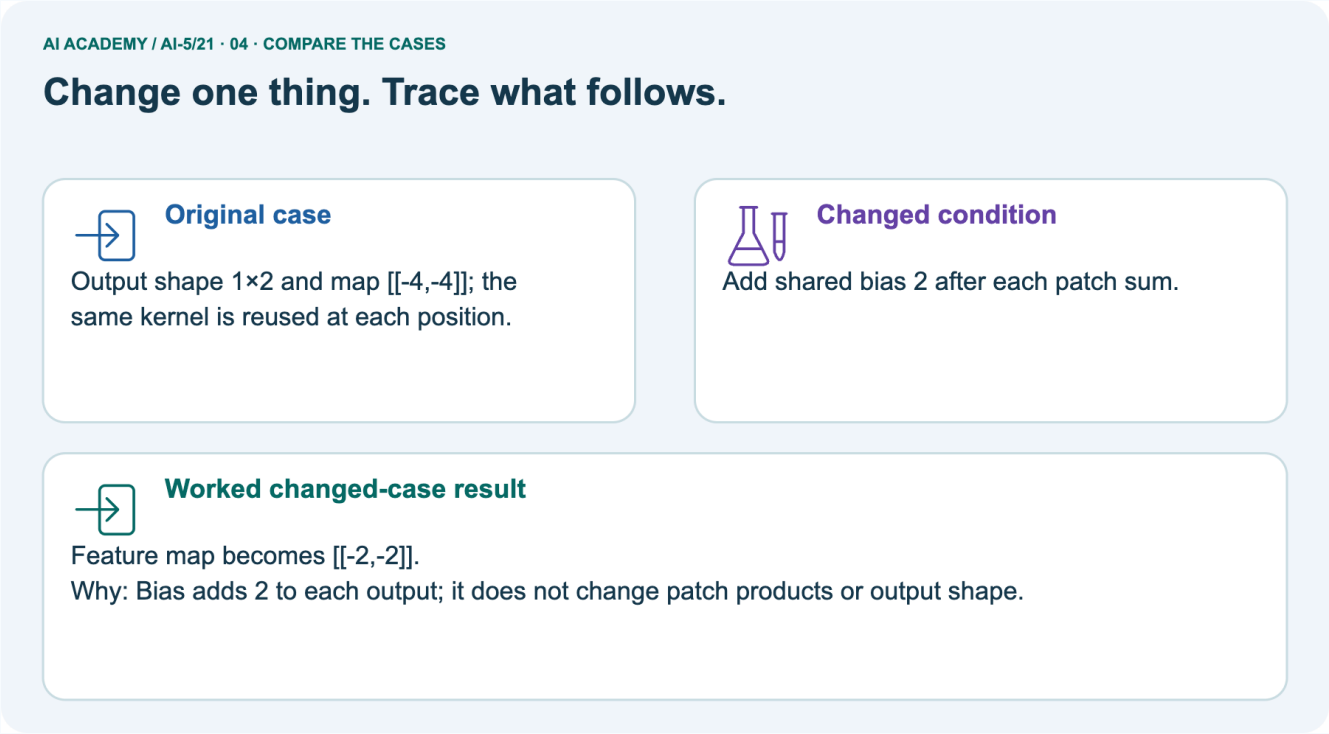
Image used by both local **kernel** windows. row 0: col 0 = 1, col 1 = 2, col 2 = 3. row 1: col 0 = 4, col 1 = 5, col 2 = 6.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Output** shape 1×2 and map [[-4,-4]]; the same **kernel** is reused at each position. Changed condition: Add shared bias 2 after each patch sum. Expected result: **Feature map** becomes [[-2,-2]].

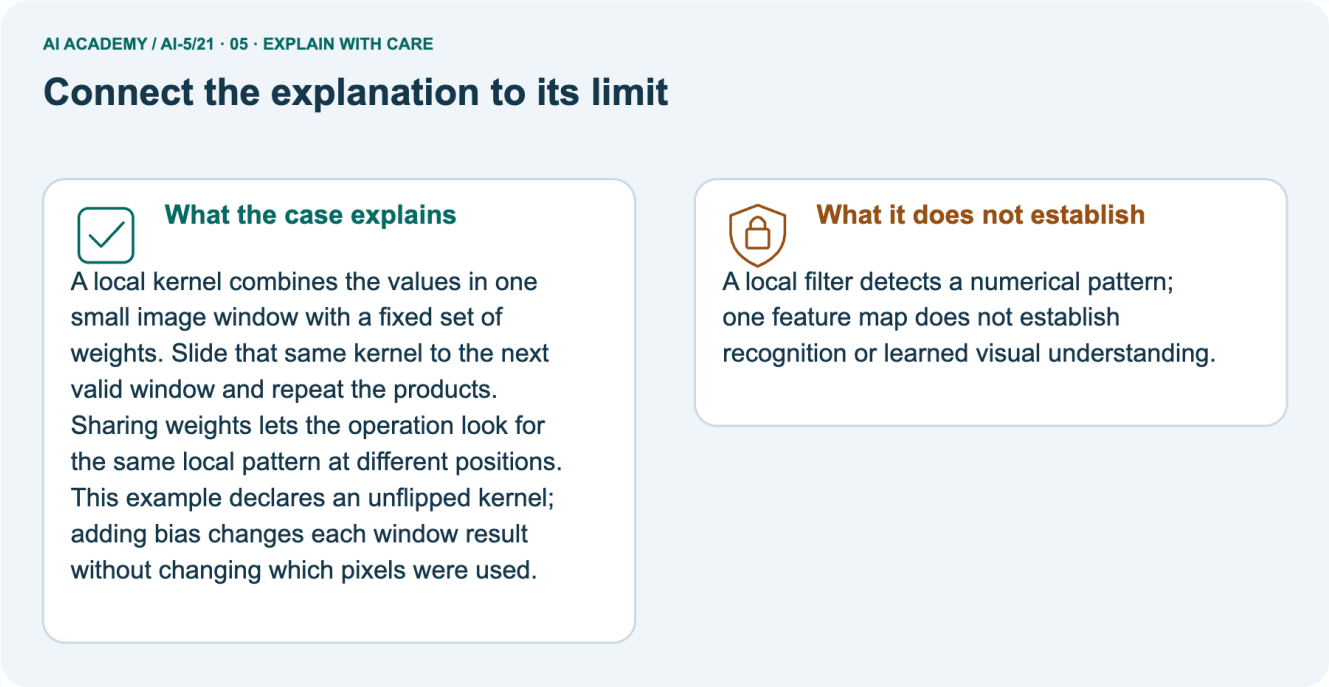
Reason: Bias adds 2 to each **output**; it does not change patch products or **output** shape.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Bias adds 2 to each **output**; it does not change patch products or **output** shape.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A local **kernel** combines the values in one small image window with a fixed set of weights. Slide that same **kernel** to the next valid window and repeat the products.

Sharing weights lets the operation look for the same local pattern at different positions. This example declares an unflipped **kernel**; adding bias changes each window result without changing which pixels were used.

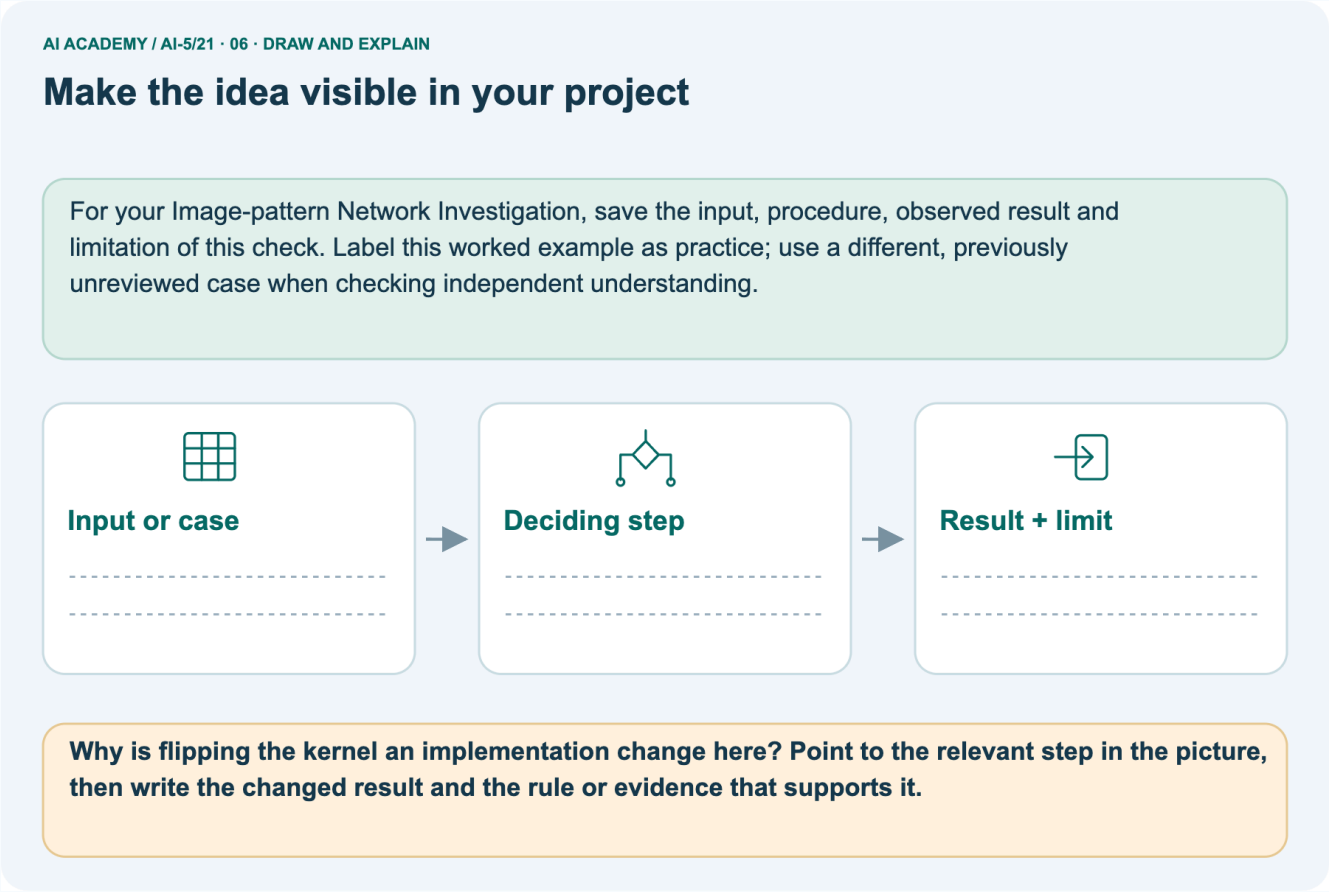
Limit: A local filter detects a numerical pattern; one **feature map** does not establish recognition or learned visual understanding.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A local filter detects a numerical pattern; one **feature map** does not establish recognition or learned visual understanding.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is flipping the **kernel** an implementation change here? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The declared operation uses an unflipped **kernel**, often called cross-correlation. Flipping it changes the arithmetic contract and may change results. Changed-case check: **Feature map** becomes [[-2,-2]].

Bias adds 2 to each **output**; it does not change patch products or **output** shape.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Convolution as a local operation: the case

02. Follow the deciding steps

03. Image used by both local kernel windows

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.